AMCS
Practice 3
Sums 1. Notation. Sums and recurrences. The repertoire and perturbation methods.

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Notation.

1. Basic Sum Notation

Formula

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + a_3 + \cdots + a_n$$

Notation

- $\sum$ — summation sign
- k — index of summation
- $k = 1$ — starting value of the index
- n — final value of the index
- a_k — term being added

How to use it

Substitute

$$k = 1, 2, 3, \ldots, n$$

into a_k and add all the resulting terms.

For example,

$$\sum_{k=1}^{4} a_k = a_1 + a_2 + a_3 + a_4.$$

Examples

1. Expand the sum:

$$\sum_{k=1}^{5} k.$$

2. Evaluate:

$$\sum_{k=1}^{4} k^2.$$

3. Evaluate:

$$\sum_{k=1}^{5} (2k + 3).$$

Evaluate each sum.

1.

$$\sum_{k=1}^{8}(2k+1)$$

2.

$$\sum_{k=1}^{6}(k^2+2)$$

3.

$$\sum_{k=2}^{7}(3k-2)$$

4.

$$\sum_{k=1}^{5}(k^2+2k)$$

9.

$$\sum_{k=1}^{5}\frac{k+1}{2}$$

10.

$$\sum_{k=0}^{5}(k^2-k+3)$$

Sum with a Condition

Formula

$$\boxed{\sum_{P(k)} a_k = \sum_{k \in K} a_k}$$

where

$$\boxed{K = \{k : P(k) \text{ is true}\}.}$$

Notation

- k — index
- $P(k)$ — a condition on k
- K — the set of all k satisfying the condition
- a_k — term being added

How to use it

First, find all values of k for which $P(k)$ is true.

Then add the corresponding terms a_k.

For example, if

$$P(k) : 1 \le k \le 10, \qquad k \text{ is even},$$

then

$$K = \{2, 4, 6, 8, 10\}.$$

Examples

1. Expand:

$$\sum_{\substack{1 \le k \le 8 \\ k \text{ even}}} k.$$

2. Evaluate:

$$\sum_{\substack{1 \le k \le 10 \\ k \text{ odd}}} k^2.$$

3. Evaluate:

$$\sum_{\substack{1 \le k \le 15 \\ 3 \mid k}} (2k + 1).$$

Here $3 \mid k$ means that k is divisible by 3.

Evaluate each sum.

1.

$$\sum_{\substack{1\le k\le 15\\ k \text{ odd}}} k$$

2.

$$\sum_{\substack{1\le k\le 16\\ k \text{ even}}} k^2$$

3.

$$\sum_{\substack{1\le k\le 20\\ 4\mid k}} (k+2)$$

4.

$$\sum_{\substack{1\le k\le 25\\ 5\mid k}} (2k-1)$$

5.

$$\sum_{\substack{5\le k\le 20\\ k \text{ odd}}} (k-2)$$

6.

$$\sum_{\substack{1\le k\le 18\\ 3\mid k}} k^2$$

7.

$$\sum_{\substack{1\le k\le 30\\ 6\mid k}} \frac{k}{2}$$

8.

$$\sum_{\substack{1\le k\le 20\\ k \text{ even}\\ k>10}} (k+1)$$

Iverson Bracket

Formula

$$[P(k)] = \begin{cases} 1, & P(k) \text{ is true,} \\ 0, & P(k) \text{ is false.} \end{cases}$$

Notation

- $P(k)$ — a condition
- $[P(k)]$ — Iverson bracket
- If $P(k)$ is true, then $[P(k)] = 1$
- If $P(k)$ is false, then $[P(k)] = 0$

Therefore,

$$a_k[P(k)]$$

is either a_k or 0.

How to use it

Consider

$$\sum_{k=1}^{6} k[k \text{ even}].$$

For odd k,

$$[k \text{ even}] = 0.$$

For even k,

$$[k \text{ even}] = 1.$$

Therefore, only the even values contribute to the sum.

Examples

1. Evaluate:

$$\sum_{k=1}^{6} k[k \text{ even}].$$

2. Evaluate:

$$\sum_{k=1}^{10} k[k > 6].$$

3. Evaluate:

$$\sum_{k=1}^{15} k[k \text{ odd}][k < 10].$$

Evaluate each sum.

1.

$$\sum_{k=1}^{12} k[k \text{ even}]$$

2.

$$\sum_{k=1}^{15} k[k \text{ odd}]$$

3.

$$\sum_{k=1}^{12} k^2[k > 8]$$

4.

$$\sum_{k=1}^{20} (k+1)[3 \mid k]$$

Sum Recurrence Formula

Formula

If

$$\boxed{S_n = \sum_{k=0}^{n} a_k,}$$

then

$$\boxed{S_0 = a_0, \qquad S_n = S_{n-1} + a_n, \quad n > 0.}$$

Notation

- S_n — the sum up to the n-th term
- S_{n-1} — the previous sum
- a_n — the new term
- n — index

Meaning

The formula says:

$$\boxed{\text{new sum} = \text{previous sum} + \text{new term}}$$

because

$$S_{n-1} = a_0 + a_1 + \cdots + a_{n-1}.$$

Therefore,

$$S_n = S_{n-1} + a_n.$$

How to use it

First, identify a_n.

Then write

$$S_n = S_{n-1} + a_n.$$

Examples

Write the corresponding recurrence relation.

1.

$$S_n = \sum_{k=0}^{n} k.$$

2.

$$S_n = \sum_{k=0}^{n} (3k + 2).$$

3.

$$S_n = \sum_{k=0}^{n} (k^2 + 2k).$$

For each sum, write the corresponding recurrence relation including the initial value.

1.

$$S_n = \sum_{k=0}^{n} (2k + 1)$$

2.

$$S_n = \sum_{k=0}^{n} (k^2 + 3)$$

3.

$$S_n = \sum_{k=0}^{n} (4k - 2)$$

4.

$$S_n = \sum_{k=0}^{n} (k^2 + 2k + 1)$$

5.

$$S_n = \sum_{k=0}^{n} 3^k$$

6.

$$S_n = \sum_{k=0}^{n} (2k^2 - k)$$

7.

$$S_n = \sum_{k=0}^{n} (5 - 2k)$$

8.

$$S_n = \sum_{k=0}^{n} (k + 2)^2$$

9.

$$S_n = \sum_{k=0}^{n} k2^k$$

10.

$$S_n = \sum_{k=0}^{n} (3k^2 + 2k - 4)$$

From Recurrence to Sum

Formula

If

$$\boxed{S_n = S_{n-1} + a_n,}$$

then the recurrence can be written as a sum.

Notation

- S_n — current value
- S_{n-1} — previous value
- a_n — term added at each step

How to use it

Identify what is added to S_{n-1}.

This expression becomes the term of the sum.

Examples

Write each recurrence as a sum.

1.

$$S_0 = 0, \qquad S_n = S_{n-1} + n.$$

2.

$$S_0 = 2, \qquad S_n = S_{n-1} + 2n.$$

3.

$$S_0 = 1, \qquad S_n = S_{n-1} + (n^2 + 1).$$

Write each recurrence as a sum.

1.

$$S_0 = 3, \qquad S_n = S_{n-1} + 2n$$

2.

$$S_0 = 1, \qquad S_n = S_{n-1} + (3n + 1)$$

3.

$$S_0 = 4, \qquad S_n = S_{n-1} + n^2$$

4.

$$S_0 = 2, \qquad S_n = S_{n-1} + (n^2 + n)$$

5.

$$S_0 = 1, \qquad S_n = S_{n-1} + 2^n$$

6.

$$S_0 = -2, \qquad S_n = S_{n-1} + (4n - 3)$$

7.

$$S_0 = 5, \qquad S_n = S_{n-1} + (n + 1)^2$$

8.

$$S_0 = 0, \qquad S_n = S_{n-1} + n3^n$$

9.

$$S_0 = 7, \qquad S_n = S_{n-1} + (2n^2 + 1)$$

10.

$$S_0 = -1, \qquad S_n = S_{n-1} + (n^2 - 2n + 3)$$

Geometric Sum

Formula

$$\sum_{k=0}^{n} a_0 q^k = a_0 \frac{1-q^{n+1}}{1-q}, \qquad q \neq 1.$$

Notation

- a_0 — first term
- q — common ratio
- k — index of summation
- n — final index
- $n+1$ — number of terms

Examples

1. Evaluate:

$$\sum_{k=0}^{4} 2^k.$$

2. Evaluate:

$$\sum_{k=0}^{5} 3\left(\frac{1}{2}\right)^k.$$

3. Evaluate:

$$\sum_{k=0}^{6} 5(-2)^k.$$

Repertoire Method

Consider a recurrence of the form

$$\boxed{R_0 = \alpha, \qquad R_n = R_{n-1} + \beta + \gamma n.}$$

Its closed formula is

$$\boxed{R_n = \alpha + n\beta + \frac{n(n+1)}{2}\gamma.}$$

Notation

- R_n — current term of the sequence
- R_{n-1} — previous term
- R_0 — initial value
- α — initial value
- β — constant term added at every step
- γ — coefficient of n
- n — index

How to use it

First, identify

$$\alpha, \qquad \beta, \qquad \gamma.$$

Then substitute them into

$$R_n = \alpha + n\beta + \frac{n(n+1)}{2}\gamma.$$

Finally, simplify the result.

Examples

Find a closed formula for R_n.

1.

$$R_0 = 2, \qquad R_n = R_{n-1} + 3.$$

2.

$$R_0 = 1, \qquad R_n = R_{n-1} + 2 + n.$$

3.

$$R_0 = 5, \qquad R_n = R_{n-1} - 3 + 4n.$$

Find a closed formula for R_n.

1. $R_0 = 4, \quad R_n = R_{n-1} + 2 + 3n$

2. $R_0 = -1, \quad R_n = R_{n-1} + 5 + 2n$

3. $R_0 = 3, \quad R_n = R_{n-1} - 2 + 4n$

4. $R_0 = 0, \quad R_n = R_{n-1} + 1 - 3n$

5. $R_0 = 6, \quad R_n = R_{n-1} - 4 + 2n$

6. $R_0 = 2, \quad R_n = R_{n-1} + 3 - 2n$

7. $R_0 = -3, \quad R_n = R_{n-1} - 1 + 5n$

8. $R_0 = 5, \quad R_n = R_{n-1} + 4 + n$

9. $R_0 = 1, \quad R_n = R_{n-1} - 5 + 6n$

10. $R_0 = 7, \quad R_n = R_{n-1} + 2 - 4n$

Repertoire Method: Derivation

Assume that the closed formula has the form

$$\boxed{R_n = A(n)\alpha + B(n)\beta + C(n)\gamma.}$$

We use three repertoire functions:

$$\boxed{R_n = 1, \qquad R_n = n, \qquad R_n = n^2.}$$

They give

$$A(n) = 1,$$

$$B(n) = n,$$

$$C(n) = \frac{n(n+1)}{2}.$$

Therefore,

$$\boxed{R_n = \alpha + n\beta + \frac{n(n+1)}{2}\gamma.}$$

How to use it

Choose simple functions for R_n and substitute them into the recurrence

$$R_n = R_{n-1} + \beta + \gamma n.$$

Use the resulting values of α, β, γ to determine

$$A(n), \qquad B(n), \qquad C(n).$$

Examples

Use the Repertoire Method to determine the closed formula.

1.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + \beta.$$

2.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + \gamma n.$$

3.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + \beta + \gamma n.$$

For each recurrence, identify α, β, γ, determine the required coefficient functions, and obtain the closed formula.

1.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + 2\beta + \gamma n$$

2.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + \beta + 2\gamma n$$

3.

$$R_0 = \alpha, \qquad R_n = R_{n-1} - \beta + \gamma n$$

4.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + 3\beta + \gamma n$$

5.

$$R_0 = \alpha, \qquad R_n = R_{n-1} + \beta - \gamma n$$

Perturbation Method

Start with

$$\boxed{S_n = \sum_{k=0}^{n} a_k.}$$

The main perturbation formula is

$$\boxed{S_n + a_{n+1} = a_0 + \sum_{k=0}^{n} a_{k+1}.}$$

Notation

- S_n — original sum
- a_k — general term
- a_0 — first term
- a_{n+1} — term after the last term
- a_{k+1} — shifted term

The main idea is

$$\boxed{\text{shift the sum} \rightarrow \text{express it in terms of } S_n \rightarrow \text{solve for } S_n.}$$

How to use it

Start with

$$S_n = \sum_{k=0}^{n} a_k.$$

Then use

$$S_n + a_{n+1} = a_0 + \sum_{k=0}^{n} a_{k+1}.$$

Transform

$$\sum_{k=0}^{n} a_{k+1}$$

so that it can be expressed in terms of S_n.

Examples

Use the Perturbation Method to find a closed formula.

1.

$$S_n = \sum_{k=0}^{n} 2^k.$$

2.

$$S_n = \sum_{k=0}^{n} 3 \cdot 2^k.$$

3.

$$S_n = \sum_{k=0}^{n} k2^k.$$

Use the Perturbation Method to find a closed formula for each sum.

1.

$$S_n = \sum_{k=0}^{n} 3^k$$

2.

$$S_n = \sum_{k=0}^{n} 4 \cdot 2^k$$

3.

$$S_n = \sum_{k=0}^{n} 5 \left(\frac{1}{2}\right)^k$$

4.

$$S_n = \sum_{k=0}^{n} (-2)^k$$

5.

$$S_n = \sum_{k=0}^{n} k3^k$$

6.

$$S_n = \sum_{k=0}^{n} k4^k$$

7.

$$S_n = \sum_{k=0}^{n} (k+1)2^k$$

8.

$$S_n = \sum_{k=0}^{n} (2k+1)2^k$$

6.

A sequence satisfies

$$R_0 = 7, \qquad R_n = R_{n-1} - 4 + 6n.$$

Do not calculate $R_1, R_2, \ldots, R_n$ one by one.

a) Identify

$$\alpha, \qquad \beta, \qquad \gamma.$$

b) Use the Repertoire Method formula to obtain a closed form for R_n.

c) Write the same sequence using summation notation.

d) Can your sum in part (c) be written with lower limit $k = 0$ while keeping the summand $-4 + 6k$? Explain.

e) Find R_{50}.

7. Reward System

A game gives a player 5 points before the first round.

After round n, the player receives

$$2 + 3n$$

additional points.

Let P_n denote the player's total score after round n.

However, points from rounds divisible by 4 are doubled.

a) Write an expression using Iverson brackets for the number of points earned in round k.

b) Write P_n using summation notation.

c) Find the total score after 8 rounds.

d) Explain why the initial 5 points should normally be written separately from the summation.

8.

Consider

$$S_n = \sum_{k=0}^{n} k3^k.$$

A student claims that this is a geometric sum and writes

$$S_n = \frac{1 - 3^{n+1}}{1 - 3}.$$

a) Explain why the student's method is incorrect.

b) Apply the Perturbation Method to the original sum.

Start from

$$S_n = \sum_{k=0}^{n} k3^k$$

and use the shifted expression to make S_n appear again.

c) Find a closed formula for S_n.

d) Use your formula to calculate

$$\sum_{k=0}^{6} k3^k.$$

10. Challenge Problem

Define

$$T_n = \sum_{k=0}^{n} (2k+1)2^k.$$

Answer the following without expanding the entire sum.

a) Find T_0.

b) Write a recurrence relation for T_n.

c) Explain why T_n is not an ordinary geometric sum.

d) Use the Perturbation Method to find a closed formula for T_n.

e) Using your closed formula, calculate T_{10}.

f) Now consider only the terms whose indices are even. Write this new sum using an Iverson bracket:

$$E_n = ?$$

Do not evaluate E_n.